

Math 121: Commutative Algebra

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Worksheet 2.1: Zariski Topology

Problem 1. Recall that the spectrum and maximal spectrum of a ring R are the sets:

$$\mathrm{Spec}(R) := \{P \subset R \mid P \text{ is a prime ideal}\}$$

and

$$\mathrm{mSpec}(R) := \{P \subset R \mid P \text{ is a maximal ideal}\}$$

Given any subset $S \subset R$ define

$$\mathbb{V}(S) := \{P \in \mathrm{Spec}(R) \mid S \subset P\}.$$

- (a) Describe $\mathrm{Spec}(\mathbb{K})$ and $\mathrm{mSpec}(\mathbb{K})$ when $\mathbb{K}$ is a field.
- (b) Describe both $\mathrm{Spec}(\mathbb{K}[x]/\langle x^3 \rangle)$ and $\mathrm{mSpec}(\mathbb{K}[x]/\langle x^3 \rangle)$ when $\mathbb{K}$ is a field.
- (c) For an arbitrary ring R , compute $\mathbb{V}(\{0\})$ and $\mathbb{V}(\{1\})$.
- (d) Explain the natural poset structure on $\mathrm{Spec}(R)$. Describe the poset for $\mathrm{Spec}(\mathbb{Z})$.

Solution.

- (a) Let $\mathbb{K}$ be a field. Then $\mathbb{K}$ only has 1 proper ideal: the ideal generated by 0. Note that this ideal is prime because $\mathbb{K}$ is an integral domain. Moreover it is also maximal because $\mathbb{K}/\langle 0 \rangle = \mathbb{K}$ is a field. So,

$$\mathrm{Spec}(\mathbb{K}) = \mathrm{mSpec}(\mathbb{K}) = \{\langle 0 \rangle\}$$

- (b) Note that the ideal generated by 0 is not prime in this ring because $x \cdot x^2 \in \langle 0 \rangle$ even though neither x nor x^2 are in $\langle x \rangle$. Then we get,

$$\mathrm{Spec}(\mathbb{K}[x]/\langle x^3 \rangle) = \mathrm{mSpec}(\mathbb{K}[x]/\langle x^3 \rangle) = \{\langle x \rangle\}$$

(c) Let R be an arbitrary ring, then,

$$\mathbb{V}(\{0\}) = \text{Spec}(R)$$

since every ideal contains 0 and,

$$\mathbb{V}(\{1\}) = \emptyset$$

since 1 is not contained in any proper ideal of R .

(d) The natural poset structure on $\text{Spec}(R)$ is given by inclusion of ideals, that is, for $P_1, P_2 \in \text{Spec}(R)$, we write $P_1 \leq P_2$ iff $P_1 \subseteq P_2$.

The poset for $\text{Spec}(\mathbb{Z})$ contains prime ideals in $\mathbb{Z}$. The zero ideal is contained in every other ideal and every other prime ideal is maximal.

A *topological space* is a set X together with a collection of subsets τ , which satisfies the following conditions: i) $\emptyset \in \tau$ and $X \in \tau$, ii) τ is closed under arbitrary unions, and iii) τ is closed under finite intersections. We call τ a *topology* on X , the elements of τ the open subsets of X , and a subset $Z \subset X$ is closed if and only if $X \setminus Z$ is open (i.e., $X \setminus Z \in \tau$). If $S \subset X$ is a subset, the *subspace topology* on S is the topology where the open sets of S are all sets of the form $S \cap U \subset S$ where $U \subset X$ is open in X . A map $f : X \rightarrow Y$ of topological spaces is *continuous* if and only if $f^{-1}(U) \subset X$ is open for all open subsets $U \subset Y$, i.e., the preimage of an open set is open.

Problem 2. The Zariski Topology.

We will now place a topology on the set $\text{Spec}(R)$, called the Zariski topology after Oscar Zariski.

- (a) Prove that if $\mathcal{S}, \mathcal{T} \subset R$ are arbitrary sets then $\mathbb{V}(\mathcal{S}) \cap \mathbb{V}(\mathcal{T}) = \mathbb{V}(\mathcal{S} \cup \mathcal{T})$. Explain why your argument extends to an *arbitrary* intersection.
- (b) Prove that if $\mathcal{S}, \mathcal{T} \subset R$ are arbitrary sets then $\mathbb{V}(\mathcal{S}) \cup \mathbb{V}(\mathcal{T}) = \mathbb{V}(\mathcal{S} \cdot \mathcal{T})$ where

$$\mathcal{S} \cdot \mathcal{T} := \{st \mid s \in \mathcal{S}, t \in \mathcal{T}\}.$$

- (c) Prove that the sets of the form $\mathbb{V}(\mathcal{S})$ are the closed sets of a topology on $\text{Spec}(R)$. We call this topology the Zariski topology on $\text{Spec}(R)$. (Hint: Rephrase the definition of a topology in terms of closed sets instead of open sets.)
- (d) Take a break and Google Oscar Zariski and read about his life.

Solution.

- (a) We prove this for arbitrary intersections. Let $\mathcal{S}_i, i \in \Lambda$ where Λ is some indexing set, be arbitrary subsets of R . Then by definition,

$$\bigcap_{i \in \Lambda} \mathbb{V}(\mathcal{S}_i) = \bigcap_{i \in \Lambda} \{[P] \in \text{Spec}(R), \mathcal{S}_i \subseteq P\} = \left\{ [P] \in \text{Spec}(R), \bigcup_{i \in \Lambda} \mathcal{S}_i \subseteq P \right\}$$

And

$$\mathbb{V}\left(\bigcup_{i \in \Lambda} \mathcal{S}_i\right) = \left\{ [P] \in \text{Spec}(R), \bigcup_{i \in \Lambda} \mathcal{S}_i \subseteq P \right\}$$

That is, $\bigcap_{i \in \Lambda} \mathbb{V}(\mathcal{S}_i) = \mathbb{V}\left(\bigcup_{i \in \Lambda} \mathcal{S}_i\right)$. Hence proved.

- (b) Let $\mathcal{S}, \mathcal{T}$ be arbitrary subsets of R . Then by definition,

$$\mathbb{V}(\mathcal{S} \cdot \mathcal{T}) = \{[P] \in \text{Spec}(R), \mathcal{S} \cdot \mathcal{T} \subseteq P\}$$

And

$$\mathbb{V}(\mathcal{S}) \cup \mathbb{V}(\mathcal{T}) = \{[P] \in \text{Spec}(R), \mathcal{S} \subseteq P\} \cup \{[P] \in \text{Spec}(R), \mathcal{T} \subseteq P\}$$

Note that $\mathcal{S} \cdot \mathcal{T} \subset \mathcal{S}, \mathcal{T}$. Then the containment $\mathbb{V}(\mathcal{S} \cdot \mathcal{T}) \subseteq \mathbb{V}(\mathcal{S}) \cup \mathbb{V}(\mathcal{T})$ is obvious.

To prove that $\mathbb{V}(\mathcal{S} \cdot \mathcal{T}) \supseteq \mathbb{V}(\mathcal{S}) \cup \mathbb{V}(\mathcal{T})$, it suffices to show that if $\mathcal{S} \cdot \mathcal{T} \subseteq P$ then either $\mathcal{S} \subseteq P$ or $\mathcal{T} \subseteq P$.

Assume $\mathcal{S} \cdot \mathcal{T} \subseteq P$. Let $s \in \mathcal{S}$ and $t \in \mathcal{T}$. Then, since P is a prime ideal and $st \in P$ then either $s \in P$ or $t \in P$. If both $s, t \in P$ we are done. Else wlog, assume that $s \notin P$. But note that $\forall t' \in \mathcal{T}, s \cdot t' \in P$. Hence, $t' \in P, \forall t' \in \mathcal{T}$, i.e. $\mathcal{T} \subseteq P$.

Hence, $\mathbb{V}(\mathcal{S}) \cup \mathbb{V}(\mathcal{T}) = \mathbb{V}(\mathcal{S} \cdot \mathcal{T})$.

(c) To prove that the sets of the form $\mathbb{V}(\mathcal{S})$ are the closed sets of the Zariski topology on $\text{Spec}(R)$ we must show that (i) $\emptyset, \text{Spec}(R)$ are closed sets, (ii) the finite union of closed sets is closed, and (iii) arbitrary intersection of closed sets is closed.

(i) Note that $\emptyset = \{\mathbb{V}(\{1\})\}$ since 1 is not contained in any proper ideal of R , and $\text{Spec}(R) = \{\mathbb{V}(\{0\})\}$ since 0 is contained in every ideal and thus in every prime ideal. Both $\mathbb{V}(\{1\})$ and $\mathbb{V}(\{0\})$ are closed sets, hence $\emptyset$ and $\text{Spec}(R)$ are closed.

(ii) From part (b) we know that the finite union of two closed sets is closed, and by induction we can extend this to a finite number of closed sets. Hence, the finite union of closed sets is closed.

(iii) From part (a) we know that any arbitrary intersection of closed sets is closed.

Hence, sets of the form $\mathbb{V}(\mathcal{S})$ are the closed sets of the Zariski topology on $\text{Spec}(R)$.

Problem 3. Describe the closed subsets of $\text{mSpec}(R) \subset \text{Spec}(R)$ in the subspace topology.

Solution.

Closed subsets of $\text{mSpec}(R)$ look like $\mathbb{V}(\mathcal{S}) \cap \text{mSpec}(R)$, for all $\mathbb{V}(\mathcal{S})$ in the Zariski Topology on $\text{Spec}(R)$. That is, the closed subsets of $\text{mSpec}(R)$ are maximal ideals M such that $\mathcal{S} \subseteq M$.

Problem 4. Let $\mathbb{K}$ be an algebraically closed field. This exercise will describe the Zariski topology on $\text{Spec}(\mathbb{K}[x])$.

- (a) Describe the points of $\text{Spec}(\mathbb{K}[x])$. (Hint: There are two types.)
- (b) Prove that if $S \subset \mathbb{K}[x]$ then $\mathbb{V}(S) = \mathbb{V}(\{f\})$ where f is the greatest common divisor of all elements of S .
- (c) Show that the subspace topology on $\text{mSpec}(\mathbb{K}[x])$ is the finite complement topology. (Recall the *finite complement topology* on a set X is defined by letting the open sets be the empty set and sets of the form $X \setminus T$ for $T \subset X$ a subset of finite cardinality.)

Solution.

- (a) Points of $\text{Spec}(\mathbb{K}[x]) = \{\langle 0 \rangle\} \cup \{\langle x - a \rangle, a \in \mathbb{K}\}$.
- (b) Note that since every element $s \in S$ is a multiple of the $\text{gcd}(S)$, the containment $\mathbb{V}(S) \subseteq \mathbb{V}(\{f\})$ is obvious. To show the opposite containment we use the following algorithm:

Require: A set S

Ensure: $\text{gcd}(S)$

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Pick  $a_1, a_2 \in S$ ; set  $d \leftarrow \text{gcd}(a_1, a_2)$ 
while there exists  $a \in S$  such that  $d \nmid a$  do
    Pick such an  $a \in S$ 
     $d \leftarrow \text{gcd}(d, a)$ 
end while
return  $d$ 

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Then, there must exist a_i, a_j such that $\text{gcd}(a_i, a_j) = d$. Then by Bezout's Identity, we have $d \in \langle S \rangle \implies \langle d \rangle \subseteq \langle p \rangle$ for all prime ideals containing S .

Hence, $\mathbb{V}(S) = \mathbb{V}(\{f\})$.

- (c) Let C be a closed set in $\text{mSpec}(\mathbb{K}[x])$. Then we know $C = \mathbb{V}(I)$ for some $I \subseteq \mathbb{K}[x]$. Since $\mathbb{K}[x]$ is a PID every ideal is of the form $I = \langle f \rangle$ for some $f \in \mathbb{K}[x]$.
 - If $f = 0$ then $\mathbb{V}(f) = \text{mSpec}(\mathbb{K}[x])$.
 - If $f \neq 0$, since $\mathbb{K}$ is algebraically closed, $f = \prod_{i=1}^n (x - a_i)$ for finitely many factors. Moreover since $\mathbb{K}[x]$ is a PID over an algebraically closed field maximal ideals of $\mathbb{K}[x]$ correspond to irreducible polynomials, which are of the form $\langle x - a \rangle$. So, $\langle f \rangle \subseteq \langle x - a_i \rangle$ for its finitely many factors.

Hence, $\mathbb{V}(I)$ is finite and the closed sets are exactly $\text{mSpec}(\mathbb{K}[x])$ itself and finite subsets, which is exactly the finite complement topology.

Problem 5. Dense Points.

If X is a topological space we say that a set $S \subset X$ is *dense* if $U \cap S \neq \emptyset$ for every non-empty open subset $U \subset X$.

- (a) Let p be a point in a topological space X . Prove that $\{p\}$ is a dense subset of X if and only if the smallest closed subset which contains $\{p\}$ is X . We call such points *dense points* of X . (Note: no point in $\mathbb{R}^n$ or $\mathbb{C}^n$ with the Euclidean topology is dense in this sense.)
- (b) Given a prime ideal $P \subset R$ we will frequently write $[P]$ to denote the corresponding point in $\text{Spec}(R)$. Prove that a point $[P] \in \text{Spec}(R)$ is dense if and only if $V(P) = \text{Spec}(R)$.
- (c) Characterize all of the dense points in the following:
 - (i) $\text{Spec}(\mathbb{K}[x])$
 - (ii) $\text{Spec}(\mathbb{Z})$
 - (iii) $\text{Spec}(\mathbb{Z}[t])$
 - (iv) $\text{Spec}(\mathbb{K}[x]/\langle x^2 \rangle)$
 - (v) $\text{Spec}(\mathbb{K}[x, y])$
 - (vi) $\text{Spec}(\mathbb{Z}/\langle 6 \rangle)$.
 - (vii) $\text{Spec}(\mathbb{Z}/\langle 4 \rangle)$.
 - (viii) $\text{Spec}(\mathbb{Z}/\langle 12 \rangle)$.
- (d) Prove that if R is a domain then $\text{Spec}(R)$ has a unique dense point.
- (e) Repeat part 5(c) replacing Spec with mSpec in the subspace topology. Can $\text{mSpec}(R)$ ever have a dense point?
- (f) Consider the following two properties (which together are implied by the Hausdorff condition) for a topological space X : i) for every $p \in X$ the set $\{p\}$ is closed and ii) given two distinct points $p, q \in X$ there exist open subsets $U, V \subset X$ such that $p \in U$ and $q \in V$ and $U \cap V = \emptyset$. Prove that if X is a topological space with a dense point, and X is not a single point, then X fails both parts of this definition.

Solution.

(a) ($\implies$)

Assume that $\{p\}$ is a dense subset of X .

Then $U \cap \{p\} \neq \emptyset$ for all non-empty open subsets $U \subset X$. That is $\{p\} \not\subseteq X/U$ (since all non-empty proper closed subsets of X are complements of non-empty open subsets of X , $\{p\}$ is not contained in any proper closed subset of X). But $\{p\} \subset X$, hence smallest closed subset which contains $\{p\}$ is X .

($\impliedby$)

Assume that the smallest closed subset which contains $\{p\}$ is X .

Then for all proper closed subsets U' of X , $\{p\} \not\subseteq U'$ that is $X/U' = U \cap \{p\} \neq \emptyset$. Note that $\emptyset \neq U$ for any U' since U' is a proper subset of X . Therefore, for every non-empty open subset $U \subset X$, $\{p\} \cap U \neq \emptyset$, and hence $\{p\}$ is dense in X .

Hence proved that $\{p\}$ is a dense subset of X if and only if the smallest closed subset which contains $\{p\}$ is X .

(b) ($\implies$)

Assume $[P] \in \text{Spec}(R)$ is dense. Then the smallest closed set containing $[P]$ is the whole space, i.e. $\text{Spec}(R)$. That is $\mathbb{V}(P) = \text{Spec}(R)$.

($\impliedby$)

Assume $\mathbb{V}(P) = \text{Spec}(R)$. Then $P \subseteq T$ for all prime ideals $T \subset R$. That is, the smallest closed set containing P is the whole space ($\text{Spec}(R)$). Hence, $[P]$ is dense in $\text{Spec}(R)$.

Hence proved that a point $[P] \in \text{Spec}(R)$ is dense if and only if $\mathbb{V}(P) = \text{Spec}(R)$.

- (c) (i) Since $\mathbb{K}[x]$ is a PID, points of $\text{Spec}(\mathbb{K}[x])$ are $\langle 0 \rangle$ and maximal ideals $\langle f \rangle$ where f is a monic irreducible polynomial. From part (b) we know that dense points of $\text{Spec}(\mathbb{K}[x])$ are points $[P] \in \text{Spec}(\mathbb{K}[x])$ such that $\mathbb{V}(P) = \text{Spec}(\mathbb{K}[x])$. Since $\mathbb{V}(f) = \{\langle f \rangle\}$, $[f]$ is not dense in $\text{Spec}(\mathbb{K}[x])$. However since $\mathbb{V}(0) = \text{Spec}(\mathbb{K}[x])$, $[0]$ is dense in $\text{Spec}(\mathbb{K}[x])$.
- (ii) Since $\mathbb{Z}$ is a PID, points of $\text{Spec}(\mathbb{Z})$ are $\langle 0 \rangle$ and maximal ideals $\langle p \rangle$ where $p \in \mathbb{Z}$ prime. From part (b) we know that dense points of $\text{Spec}(\mathbb{Z})$ are points $[P] \in \text{Spec}(\mathbb{Z})$ such that $\mathbb{V}(P) = \text{Spec}(\mathbb{Z})$. Since $\mathbb{V}(p) = \langle p \rangle$, $[p]$ is not dense in $\text{Spec}(\mathbb{Z})$. However since $\mathbb{V}(0) = \text{Spec}(\mathbb{Z})$, $[0]$ is dense in $\text{Spec}(\mathbb{Z})$.
- (iii) From part (b) we know that dense points of $\text{Spec}(\mathbb{Z}[t])$ are points $[P] \in \text{Spec}(\mathbb{Z}[t])$ such that $\mathbb{V}(P) = \text{Spec}(\mathbb{Z}[t])$. Since $\mathbb{V}(0) = \text{Spec}(\mathbb{Z}[t])$, $[0]$ is dense in $\text{Spec}(\mathbb{Z}[t])$, and for every non-zero $[P] \in \text{Spec}(\mathbb{Z}[t])$, $\mathbb{V}(P)$ doesn't contain $[0]$ since $\langle P \rangle \not\subseteq \langle 0 \rangle$. Hence, $[P]$ is not dense.
- (iv) $\text{Spec}(\mathbb{K}[x]/\langle x^2 \rangle) = \{\langle x \rangle\}$ since $\langle 0 \rangle$ is not prime. Hence $[x]$ is trivially dense in $\text{Spec}(\mathbb{K}[x]/\langle x^2 \rangle)$.
- (v) From part (b) we know that dense points of $\text{Spec}(\mathbb{K}[x, y])$ are points $[P] \in \text{Spec}(\mathbb{K}[x, y])$ such that $\mathbb{V}(P) = \text{Spec}(\mathbb{K}[x, y])$. Since $\mathbb{V}(0) = \text{Spec}(\mathbb{K}[x, y])$, $[0]$ is dense, and for every non-zero $[P] \in \text{Spec}(\mathbb{K}[x, y])$, $\mathbb{V}(P)$ doesn't contain $[0]$ since $\langle P \rangle \not\subseteq \langle 0 \rangle$. Hence, $[P]$ is not dense.
- (vi) $\text{Spec}(\mathbb{Z}/\langle 6 \rangle) = \{\langle 2 \rangle, \langle 3 \rangle\}$. Note that they are both maximal since $\mathbb{Z}/\langle 6 \rangle/\langle 2 \rangle = \mathbb{Z}/\langle 2 \rangle$ and $\mathbb{Z}/\langle 6 \rangle/\langle 3 \rangle = \mathbb{Z}/\langle 3 \rangle$. From part (b) we know that dense points of $\text{Spec}(\mathbb{Z}/\langle 6 \rangle)$ are points $[P] \in \text{Spec}(\mathbb{Z}/\langle 6 \rangle)$ such that $\mathbb{V}(P) = \text{Spec}(\mathbb{Z}/\langle 6 \rangle)$. Since both ideals are maximal

and incomparable, $\mathbb{V}([2])$ doesn't contain $[3]$ and $\mathbb{V}([3])$ doesn't contain $[2]$. Hence, there are no dense points in $\text{Spec}(\mathbb{Z}/\langle 6 \rangle)$.

(vii) $\text{Spec}(\mathbb{Z}/\langle 4 \rangle) = \{\langle 2 \rangle\}$. Hence $[2]$ is trivially dense.

(viii) $\text{Spec}(\mathbb{Z}/\langle 12 \rangle) = \{\langle 2 \rangle, \langle 3 \rangle\}$. Note that they are both maximal since $\mathbb{Z}/\langle 12 \rangle / \langle 2 \rangle = \mathbb{Z}/\langle 2 \rangle$ and $\mathbb{Z}/\langle 12 \rangle / \langle 3 \rangle = \mathbb{Z}/\langle 3 \rangle$.

From part (b) we know that dense points of $\text{Spec}(\mathbb{Z}/\langle 12 \rangle)$ are points $[P] \in \text{Spec}(\mathbb{Z}/\langle 12 \rangle)$ such that $\mathbb{V}(P) = \text{Spec}(\mathbb{Z}/\langle 12 \rangle)$. Since both ideals are maximal and incomparable, $\mathbb{V}([2])$ doesn't contain $[3]$ and $\mathbb{V}([3])$ doesn't contain $[2]$. Hence, there are no dense points in $\text{Spec}(\mathbb{Z}/\langle 12 \rangle)$.

(d) Assume that R is an integral domain.

Existence: Note that $\langle 0 \rangle \subseteq R$ is a prime ideal, and also $\langle 0 \rangle \subseteq P$ for all prime ideals P in R . Also recall that the smallest closed set containing $[0]$ is $\mathbb{V}([0])$, and

$$\mathbb{V}([0]) = \mathbb{V}(\langle 0 \rangle) = \text{Spec}(R)$$

That is the smallest closed set containing $[0]$ is the whole space, hence $[0]$ is a dense point in $\text{Spec}(R)$.

Uniqueness: Suppose $[P] \in \text{Spec}(R)$ is also a dense point. Then by part(b), we know that $\mathbb{V}([P]) = \text{Spec}(R)$. That is $P \subseteq \langle 0 \rangle$, and $\langle 0 \rangle \subseteq P$ trivially. Hence, $[P] = [0]$.

Hence if R is a domain, $\text{Spec}(R)$ has a unique dense point.

- (e) (i) Since $\mathbb{K}[x]$ is a PID, points of $\text{mSpec}(\mathbb{K}[x])$ are maximal ideals $\langle f \rangle$ where f is a monic irreducible polynomial. Since $\mathbb{V}(f) = \langle f \rangle$, $[f]$ is not dense in $\text{mSpec}(\mathbb{K}[x])$.
- (ii) Since $\mathbb{Z}$ is a PID, points of $\text{mSpec}(\mathbb{Z})$ are maximal ideals $\langle p \rangle$ where $p \in \mathbb{Z}$ prime. Since $\mathbb{V}(p) = \langle p \rangle$, $[p]$ is not dense in $\text{Spec}(\mathbb{Z})$.
- (iii) For every non-zero $[P] \in \text{mSpec}(\mathbb{Z}[t])$, $\mathbb{V}(P) = \{[P]\}$. Hence, $[P]$ is not dense.
- (iv) $\text{mSpec}(\mathbb{K}[x]/\langle x^2 \rangle) = \{\langle x \rangle\}$. Hence $[x]$ is trivially dense in $\text{mSpec}(\mathbb{K}[x]/\langle x^2 \rangle)$.
- (v) For every maximal ideal $[P] \in \text{mSpec}(\mathbb{K}[x, y])$, $\mathbb{V}(P) = \{[P]\} \neq \text{mSpec}(\mathbb{K}[x, y])$. Hence, $[P]$ is not dense.
- (vi) $\text{mSpec}(\mathbb{Z}/\langle 6 \rangle) = \{\langle 2 \rangle, \langle 3 \rangle\}$. Since $\mathbb{V}([2])$ doesn't contain $[3]$ and $\mathbb{V}([3])$ doesn't contain $[2]$ there are no dense points in $\text{mSpec}(\mathbb{Z}/\langle 6 \rangle)$.
- (vii) $\text{mSpec}(\mathbb{Z}/\langle 4 \rangle) = \{\langle 2 \rangle\}$. Hence $[2]$ is trivially dense.
- (viii) $\text{mSpec}(\mathbb{Z}/\langle 12 \rangle) = \{\langle 2 \rangle, \langle 3 \rangle\}$. Since $\mathbb{V}([2])$ doesn't contain $[3]$ and $\mathbb{V}([3])$ doesn't contain $[2]$ there are no dense points in $\text{mSpec}(\mathbb{Z}/\langle 12 \rangle)$.
- $\text{mSpec}(R)$ only has dense points when it is a singleton set.

(f) Assume X is a topological space with a dense point p and X is not a single point.

Then by definition of denseness $p \in U$, for all non-empty open subsets $U \subset X$, i.e. we cannot find two distinct open subsets V, W such that $V \cap W = \emptyset$.

For the sake of contradiction assume $\{p\}$ is a closed set, then $X \setminus \{p\}$ is a non-empty open set (since X is not a single point) not containing p contradicting the definition of denseness. Hence, $\{p\}$ cannot be closed.

Problem 6. Let $\mathcal{S} \subset R$ be any subset. Prove that $\mathbb{V}(\mathcal{S}) = \mathbb{V}(\langle \mathcal{S} \rangle)$ where $\langle \mathcal{S} \rangle$ denotes the ideal generated by $\mathcal{S}$.

Solution.

Let I be an ideal containing $\mathcal{S}$, then one can see that $\langle \mathcal{S} \rangle \subseteq I$. And trivially $\mathcal{S} \subset \langle \mathcal{S} \rangle \subseteq I$. Hence,

$$\mathcal{S} \subseteq I \iff \langle \mathcal{S} \rangle \subseteq I$$

And it follows that $\mathbb{V}(\mathcal{S}) = \mathbb{V}(\langle \mathcal{S} \rangle)$.

Problem 7. In this exercise we wish to understand the Zariski topology on $\text{Spec}(\mathbb{C}[x, y])$. Fix a point $p = (a, b) \in \mathbb{C}^2$.

- (a) Use the *evaluation-at- p* map: $\text{ev} : \mathbb{C}[x, y] \rightarrow \mathbb{C}$ to prove that the set of polynomial functions vanishing at p is a maximal ideal $\mathfrak{m}_p \subset \mathbb{C}[x, y]$.
- (b) Find generators for the ideal $\mathfrak{m}_p$. Is this ideal principal?
- (c) Prove that the following map is injective:

$$\begin{array}{ccc} \mathbb{C}^2 & \xrightarrow{\eta} & \text{Spec}(\mathbb{C}[x, y]) \\ p & \longmapsto & \mathfrak{m}_p \end{array}.$$

- (d) Show that the map η from part 3c is not surjective by finding at least two distinct points in $\text{Spec}(\mathbb{C}[x, y])$ not in the image of η .
- (e) Describe the closed subsets of $\text{Spec}(\mathbb{C}[x, y])$. (Hint: They come in four types)

Solution.

- (a) We have, $\text{ev}_p : \mathbb{C}[x, y] \rightarrow \mathbb{C}$ where $f \mapsto f(a, b)$. If f vanishes at p then $f(a, b) = 0$, i.e. $f \in \ker(\text{ev}_p) = \mathfrak{m}_p$. The field homomorphism is surjective (constant functions), hence $\mathfrak{m}_p$ is maximal (First Isomorphism Theorem).
- (b) Note that $\mathfrak{m}_p \supseteq \langle x - a, y - b \rangle$ since both $x - a$ and $y - b$ vanish at (a, b) . Moreover since both $\langle x - a, y - b \rangle$ ($\mathbb{C}[x, y] / \langle x - a, y - b \rangle \cong \mathbb{C}$) and $\mathfrak{m}_p$ are maximal they must be equal.
- (c) Let $p = (a, b), q = (c, d) \in \mathbb{C}^2$. Assume $\eta(p) = \eta(q)$, then

$$\mathfrak{m}_p = \mathfrak{m}_q \implies x - a \in \langle x - c, y - d \rangle \implies c = a$$

similarly $y - b \in \langle x - c, y - d \rangle \implies b = d$. Hence $p = q$.

- (d) $\langle x \rangle, \langle y \rangle \in \text{Spec}(\mathbb{C}[x, y])$ but not in $\text{im}(\eta)$ since the latter only consists of maximal ideals.
- (e) The closed subsets of $\text{Spec}(\mathbb{C}[x, y])$ are of finite unions and arbitrary intersections of the following:

- $\mathbb{V}(\langle 1 \rangle) = \emptyset$
- $\mathbb{V}(\langle 0 \rangle) = \text{Spec}(\mathbb{C}[x, y])$

- $\mathbb{V}(\langle f \rangle) = \{[\mathfrak{p}] \in \text{Spec}(\mathbb{C}[x, y]), \langle f \rangle \subseteq \mathfrak{p}\}$, f is an irreducible polynomial.
- $\mathbb{V}(\mathfrak{m}_p) = \{\mathfrak{m}_p\}$, where $\mathfrak{m}_p$ is the maximal ideal that corresponds to the kernel of evaluation-at- p map.

A function $f : X \rightarrow Y$ of topological spaces is *continuous* if and only if $f^{-1}(U) \subset X$ is open for all open subsets $U \subset Y$, i.e., the preimage of an open is open. A *homeomorphism* $f : X \rightarrow Y$ is a continuous map of topological spaces with a continuous inverse $f^{-1} : Y \rightarrow X$. A useful fact is that a continuous map $f : X \rightarrow Y$ is a homeomorphism if and only if it is bijective and $f(U) \subset Y$ is open for every open subsets $U \subset X$.

Problem 8. Ring Maps Induce Continuous Maps.

Let $\phi : R \rightarrow S$ be a ring homomorphism. The goal of this exercise is to show ϕ induces a map $\phi^\# : \text{Spec}(S) \rightarrow \text{Spec}(R)$, which is continuous in the Zariski topology.

- (a) Prove that if $[P] \in \text{Spec}(S)$, then $[\phi^{-1}(P)] \in \text{Spec}(R)$.
- (b) Prove that the map below is continuous in the Zariski topology:

$$\begin{array}{ccc} \text{Spec}(S) & \xrightarrow{\phi^\#} & \text{Spec}(R) \\ [P] & \longmapsto & [\phi^{-1}(P)] \end{array}$$

- (c) For the ring map $\mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z}$ sending each integer to its residue class modulo 2, describe the induced map on spectra explicitly.
- (d) For the ring map $\mathbb{Z} \hookrightarrow \mathbb{Q}$, describe the induced map on spectra explicitly.
- (e) If $\phi : R \hookrightarrow S$ is an injective ring map, give a simple description of the induced map on spectra.

Solution.

- (a) We know that pre-images of ideals are ideals, so we only need to show that $\phi^{-1}(P)$ is prime. Let $ab \in P$, then $\phi^{-1}(ab) \in \phi^{-1}(P)$. Since P is prime either $a \in P$ or $b \in P$, and hence either $\phi^{-1}(a) \in \phi^{-1}(P)$ or $\phi^{-1}(b) \in \phi^{-1}(P)$. Hence $\phi^{-1}(P)$ is prime, and $[\phi^{-1}(P)] \in \text{Spec}(R)$.

- (b) We will show that pre-images of closed sets in $\text{Spec}(R)$ are closed. Let $\mathbb{V}(I)$ be a closed set for some ideal $I \subseteq R$. Then,

$$(\phi^\#)^{-1}(\mathbb{V}(I)) = \{\mathfrak{p} \mid I \subseteq \phi^{-1}(\mathfrak{p})\} = \{\mathfrak{p} \mid \langle \phi(I) \rangle \subseteq \mathfrak{p}\} = \mathbb{V}(\langle \phi(I) \rangle)$$

Which is a closed set in $\text{Spec}(S)$.

- (c) The map induced on spectra is $\text{Spec}(\mathbb{Z}/2\mathbb{Z}) \xrightarrow{\phi^\#} \text{Spec}(\mathbb{Z})$. $[0]$ is the only element in $\text{Spec}(\mathbb{Z}/2\mathbb{Z})$ and $\phi^\#([0]) = [2] \in \text{Spec}(\mathbb{Z})$.
- (d) The map induced on spectra is $\text{Spec}(\mathbb{Q}) \xrightarrow{\phi^\#} \text{Spec}(\mathbb{Z})$. $[0]$ is the only element in $\text{Spec}(\mathbb{Q})$ and $\phi^\#([0]) = [0] \in \text{Spec}(\mathbb{Z})$.
- (e) If $\phi : R \hookrightarrow S$ is injective, we can view R as a subring of S and $\phi^\#([\mathfrak{p}]) = [\phi^{-1}(\mathfrak{p})] = [\mathfrak{p} \cap R] \in \text{Spec}(R)$.

A *category* is a collection of objects together with, for each pair of objects, a set of morphisms between them, equipped with an associative composition law and an identity morphism for each object. A *functor* between categories $\mathbf{C}$ and $\mathbf{D}$ is a mapping that assigns to each object of $\mathbf{C}$ an object of $\mathbf{D}$ and to each morphism in $\mathbf{C}$ a morphism in $\mathbf{D}$, preserving identity morphisms and composition. A functor is *contravariant* if it reverses the order of composition and *covariant* if it preserves the order of composition.

Problem 9. Spec as a Functor.

In this problem we package together much of this worksheet to define the Spec functor. Consider the map:

$$\begin{array}{ccc} \mathbf{comRing} & \longrightarrow & \mathbf{Top} \\ R & \longmapsto & \mathrm{Spec}(R) \end{array}$$

from the category of commutative rings with unit with ring homomorphisms to the category of topological spaces with continuous maps.

- (a) Explain using Question 8 how this mapping defines a contravariant functor from **comRing** to **Top**. To check functoriality, verify

$$(\mathrm{Id}_R)^\# = \mathrm{Id}_{\mathrm{Spec}(R)} \quad \text{and} \quad (\psi \circ \phi)^\# = \phi^\# \circ \psi^\#$$

for ring maps $\phi : R \rightarrow S$ and $\psi : S \rightarrow T$. We call this the Spec functor.

- (b) Do isomorphic rings have Spectra that are homeomorphic topological spaces? Explain.
- (c) If two rings have homeomorphic spectra, are they necessarily isomorphic rings? Explain.

Solution.

- (a) We already showed that a homomorphism between rings $R \rightarrow S$ corresponds to a map between their spectra $\mathrm{Spec}(S) \rightarrow \mathrm{Spec}(R)$, and that this map is a continuous topological map. So it only remains to check that the functorial properties hold,

- Let $\mathrm{id}_R : R \rightarrow R$ be the identity map. Then for $[\mathfrak{p}] \in \mathrm{Spec}(R)$,

$$(\mathrm{id}_R)^\#([\mathfrak{p}]) = [\mathrm{id}_R^{-1}(\mathfrak{p})] = [\mathfrak{p}] = \mathrm{id}_{\mathrm{Spec}(R)}([\mathfrak{p}])$$

- Let $\phi : R \rightarrow S$, $\psi : S \rightarrow T$ be two ring homomorphisms. Then for $[\mathfrak{p}] \in \mathrm{Spec}(T)$,

$$\begin{aligned} (\psi \circ \phi)^\#([\mathfrak{p}]) &= [(\psi \circ \phi)^{-1}(\mathfrak{p})] = [\phi^{-1}(\psi^{-1}(\mathfrak{p}))] \\ &= [\phi^{-1}(\psi^\#(\mathfrak{p}))] = [\phi^\#(\psi^\#(\mathfrak{p}))] \\ &= [\phi^\# \circ \psi^\#(\mathfrak{p})] \end{aligned}$$

Hence $(\mathrm{Id}_R)^\# = \mathrm{Id}_{\mathrm{Spec}(R)}$ and $(\psi \circ \phi)^\# = \phi^\# \circ \psi^\#$. The reversal of order makes this a covariant functor.

- (b) If $\phi : R \xrightarrow{\sim} S$ is a ring isomorphism then $\phi^{-1} : S \xrightarrow{\sim} R$ is also a ring isomorphism. Then,

$$(\phi^{-1} \circ \phi)^{\#} = \phi^{\#} \circ (\phi^{-1})^{\#} = (\text{Id}_R)^{\#} = \text{Id}_{\text{Spec}(R)}$$

$$(\phi \circ \phi^{-1})^{\#} = (\phi^{-1})^{\#} \circ \phi^{\#} = (\text{Id}_S)^{\#} = \text{Id}_{\text{Spec}(S)}$$

so $\phi^{\#}$ and $(\phi^{-1})^{\#}$ are inverse to each other. Since both $\phi^{\#}$ and $(\phi^{-1})^{\#}$ are continuous, $\phi^{\#} : \text{Spec}(S) \rightarrow \text{Spec}(R)$ is a homeomorphism. Hence isomorphic rings have homeomorphic spectra.

- (c) No, because we could have a homeomorphism between spectra of fields ($\text{Spec}(\mathbb{C}) \cong \text{Spec}(\mathbb{R})$) but that doesn't necessarily correspond to an isomorphism of fields ($\mathbb{C} \not\cong \mathbb{R}$).

Problem 10. Find a homeomorphism between $\text{Spec}(\mathbb{C}[x])$ and $\text{Spec}(\mathbb{C}[x, y]/\langle x - y \rangle)$. Construct the corresponding ring map explicitly in *Macaulay2* and compute its kernel.

Solution. Since $\mathbb{C}[x, y]/\langle x - y \rangle \cong \mathbb{C}[x]$ we send $\mathfrak{p} \subseteq \mathbb{C}[x]$ to $\bar{\mathfrak{p}} \subseteq \mathbb{C}[x, y]/\langle x - y \rangle$. And the corresponding ring map is $\mathbb{C}[x, y]/\langle x - y \rangle \rightarrow \mathbb{C}[x]$ with trivial kernel.

Problem 11. Closed Embeddings I.

Let $I \subset R$ be an ideal and $\pi : R \rightarrow R/I$ the natural quotient map. Consider the induced map on spectra

$$\operatorname{Spec}(R/I) \xrightarrow{\pi^\#} \operatorname{Spec}(R)$$

In this exercise we will show this map is a closed embedding. A *closed embedding* is a map of topological spaces whose image is closed and which is a homeomorphism onto its image.

- (a) Using the correspondence between ideals in R and R/I give an explicit description of $\pi^\#$.
- (b) Prove that the image of $\pi^\#$ is equal to $\mathbb{V}(I)$.
- (c) Explain why the ideal correspondence implies that $\pi^\#$ is injective.
- (d) Prove that $\pi^\#$ is a closed embedding onto $\mathbb{V}(I)$, in the subspace topology, by showing that if $J \subset R/I$ is an ideal then

$$\pi^\#(\mathbb{V}(J)) = \mathbb{V}(\pi^{-1}(J)).$$

Solution.

- (a) Ideals in R/I correspond to ideals in R containing I . And for any ideal $I \subseteq R$ prime ideal we have $\pi(I) = \bar{I}$. Let $\bar{\mathfrak{p}} \in \operatorname{Spec}(R/I)$. Then $\pi^\#(\bar{\mathfrak{p}}) = \pi^{-1}(\bar{\mathfrak{p}}) = \mathfrak{p}$.
- (b) The image of $\pi^\#$ by definition is

$$\operatorname{im}(\pi^\#) = \pi^\#(\operatorname{Spec}(R/I)) = \{\mathfrak{p} \in \operatorname{Spec}(R) \mid I \subseteq \pi(\mathfrak{p})\} = \mathbb{V}(I)$$

- (c) Suppose $\pi^\#(\bar{\mathfrak{p}}) = \pi^\#(\bar{\mathfrak{q}})$ then,

$$\pi^{-1}(\bar{\mathfrak{p}}) = \pi^{-1}(\bar{\mathfrak{q}}) \implies \mathfrak{p} = \mathfrak{q}$$

since π is surjective.

- (d) If $J \subseteq R/I$ is an ideal then $I \subseteq \pi^{-1}(J)$

$$\pi^\#(\mathbb{V}(J)) = \{\pi^\#(\bar{\mathfrak{p}}) \mid J \subseteq \bar{\mathfrak{p}}\} = \{\pi^{-1}(\bar{\mathfrak{p}}) \mid \pi^{-1}(J) \subseteq \pi^{-1}(\bar{\mathfrak{p}})\} = \mathbb{V}(\pi^{-1}(J))$$

Since $\pi^\#$ sends closed sets to closed sets and is a bijection onto $\mathbb{V}(I)$, it is a homeomorphism onto its image, and hence a closed embedding.